1. SYSTEM OVERVIEW

🧠 Project Title

Smart Job & Freelance Ecosystem with Secure Payments, Alerts, Screening, Reputation & Commission System

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🧑‍🔧 2. ROLE SYSTEM (FINAL)

🧑 USER (Single Account System)

A single user can act as:

🔷 Employer

- Post jobs
- Manage applicants
- Conduct screening/interviews
- Hire candidates

🔷 Job Seeker

- Browse jobs
- Apply for jobs
- Receive alerts
- Track applications

🔷 Freelancer

- Create gigs & packages
- Deliver work
- Receive payments

🔷 Client

- Order gigs
- Pay securely
- Request revisions
- Review freelancers

🛠️ ADMIN

- Manage users
- Handle disputes & refunds
- Monitor transactions
- Control commission
- View analytics

🗂️ 3. FINAL DATABASE DESIGN (WITH PK & FK)

🔐 USERS & AUTH SYSTEM

users

- id (PK)
- name
- email (UNIQUE)
- password_hash
- phone
- city
- area
- profile_image
- is_verified
- status (active/suspended)
- created_at

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user_sessions

- id (PK)
- user_id (FK → users.id)
- jwt_token
- device_info

- ip_address
- expires_at

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password_resets

- id (PK)
- user_id (FK → users.id)
- reset_token
- expires_at

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user_verifications

- id (PK)
- user_id (FK → users.id)
- otp_code
- type (email/phone)
- expires_at
- is_verified

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user_skills

- id (PK)
- user_id (FK → users.id)
- skill_name

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resumes

- id (PK)
- user_id (FK → users.id)
- summary
- education
- experience
- certifications
- github_link
- portfolio_link

💼 JOB SYSTEM

jobs

- id (PK)
- employer_id (FK → users.id)
- title
- description
- city
- area
- salary_min
- salary_max
- job_type
- experience_required
- deadline
- status (active/closed)
- created_at

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job_skills

- id (PK)
- job_id (FK → jobs.id)
- skill_name

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job_applications

- id (PK)
- job_id (FK → jobs.id)
- applicant_id (FK → users.id)
- cover_letter
- status (applied/shortlisted/interview/selected/rejected)
- applied_at

🎯 SCREENING & HIRING

hiring_pipeline

- id (PK)
- job_id (FK → jobs.id)
- applicant_id (FK → users.id)
- stage (applied → screened → interviewed → selected → hired)
- updated_at

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interview_rounds

- id (PK)
- job_id (FK → jobs.id)
- applicant_id (FK → users.id)
- round_no
- type (screening/interview/task)
- status (pending/passed/failed)
- feedback
- scheduled_at

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job_shortlists

- id (PK)
- job_id (FK → jobs.id)
- applicant_id (FK → users.id)
- notes

🔔 JOB ALERTS

job_alerts

- id (PK)
- user_id (FK → users.id)
- keyword
- skill_filter
- location_filter
- salary_min
- salary_max

💻 FREELANCE SYSTEM

gigs

- id (PK)
- freelancer_id (FK → users.id)
- title
- description
- category
- status

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gig_packages

- id (PK)
- gig_id (FK → gigs.id)
- name (Basic/Standard/Premium)
- price
- delivery_days
- features

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orders

- id (PK)
- gig_id (FK → gigs.id)
- client_id (FK → users.id)
- freelancer_id (FK → users.id)
- total_amount
- commission_amount
- status (pending/active/submitted/revision/completed/disputed)
- created_at
- completed_at

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order_revisions

- id (PK)

- order_id (FK → orders.id)
- requested_by (FK → users.id)
- message
- created_at

💬 CHAT SYSTEM

chats

- id (PK)
- order_id (FK → orders.id)
- is_active
- start_time
- expiry_time

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messages

- id (PK)
- chat_id (FK → chats.id)
- sender_id (FK → users.id)
- message
- status (sent/delivered/seen)
- created_at

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files

- id (PK)
- message_id (FK → messages.id)
- file_url
- file_type

💳 PAYMENT SYSTEM

wallet

- id (PK)
- user_id (FK → users.id)
- balance

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payment_methods

- id (PK)
- user_id (FK → users.id)
- type (bank/card)
- masked_details

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payments

- id (PK)
- user_id (FK → users.id)
- order_id (FK → orders.id)
- amount
- method (wallet/bank/card)
- status (pending/success/failed)

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transactions

- id (PK)
- user_id (FK → users.id)
- order_id (FK → orders.id)
- amount
- type (payment/payout/commission)
- platform_fee
- freelancer_amount
- previous_hash
- current_hash
- created_at

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platform_earnings

- id (PK)

- order_id (FK → orders.id)
- commission_amount

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withdrawals

- id (PK)
- user_id (FK → users.id)
- amount
- method
- status

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refunds

- id (PK)
- order_id (FK → orders.id)
- amount
- reason
- status

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transaction_logs

- id (PK)
- user_id (FK → users.id)
- action
- amount
- ip_address
- device_info

☆ RATING & REPUTATION

ratings

- id (PK)
- order_id (FK → orders.id)
- reviewer_id (FK → users.id)

- reviewed_user_id (FK → users.id)
- rating
- comment

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user_reputation

- user_id (PK, FK → users.id)
- score
- completed_jobs
- completed_orders
- disputes

🏆 LEVEL & BADGES

user_levels

- user_id (PK, FK → users.id)
- freelancer_level
- employer_level

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badges

- id (PK)
- name

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user_badges

- id (PK)
- user_id (FK → users.id)
- badge_id (FK → badges.id)

⚠️ DISPUTE SYSTEM

disputes

- id (PK)
- order_id (FK → orders.id)
- raised_by (FK → users.id)
- reason
- status (open/in_review/resolved)
- admin_action

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dispute_evidence

- id (PK)
- dispute_id (FK → disputes.id)
- file_url
- uploaded_by (FK → users.id)

🔍 SEARCH SYSTEM

search_history

- id (PK)
- user_id (FK → users.id)
- keyword
- created_at

🔔 NOTIFICATIONS

notifications

- id (PK)
- user_id (FK → users.id)
- message
- type
- is_read

📊 ANALYTICS

analytics_daily

- id (PK)
- date
- total_users
- total_jobs
- total_gigs
- total_orders
- revenue
- commission

🔄 4. COMPLETE WORKFLOWS

💼 JOB WORKFLOW

1. Employer posts job → jobs
2. System matches & sends alerts → job_alerts
3. Job seeker applies → job_applications
4. Employer shortlists → job_shortlists
5. Screening rounds → interview_rounds
6. Pipeline update → hiring_pipeline
7. Candidate selected → job closed
8. Rating → reputation update

💻 FREELANCE WORKFLOW

1. Freelancer creates gig + packages
2. Client orders gig → orders
3. Payment processed → payments
4. Transaction hash created → transactions
5. Chat activated → chats/messages
6. Work delivered → files
7. Client:

    - approve → complete
    - request revision → order_revisions
    - dispute → disputes
8. Payment split:
    - admin commission
    - freelancer earnings
9. Ratings → reputation + level update

💬 CHAT WORKFLOW

- Active only during order
- File sharing allowed
- After completion:
    - extend 30 days OR close
- Auto expire

💰 PAYMENT WORKFLOW

1. User selects method
2. Payment recorded
3. Transaction hash generated
4. Stored in transactions
5. Escrow holds funds
6. After approval:
    - commission deducted
    - payout released

🔐 SECURITY WORKFLOW

- Password → BCrypt hashing
- JWT → authentication
- Transactions → hash chain
- Card → masked
- Logs → recorded

🏆 RANKING SYSTEM

Score =

- rating (40%)
- completion (30%)
- dispute (-)
- activity (15%)
- success rate (15%)

📈 LEVEL SYSTEM

Freelancer:

Beginner → Junior → Skilled → Expert → Top Rated

Employer:

New → Active → Verified → Enterprise

📊 ANALYTICS

Admin dashboard shows:

- revenue
- commission
- user growth
- job trends
- top performers

🎯 FINAL RESULT

✓ Job portal (alerts + screening + hiring pipeline)
✓ Freelance marketplace (gig + orders)

✓ Secure payment (wallet + escrow + commission)
✓ Chat system (controlled + file sharing)
✓ Reputation system (rating + ranking + levels + badges)
✓ Dispute & refund system
✓ Search + analytics